

Lattice : A partially ordered set in which every pair of elements has both a least upper bound and a greatest lower bound is called a **lattice**.

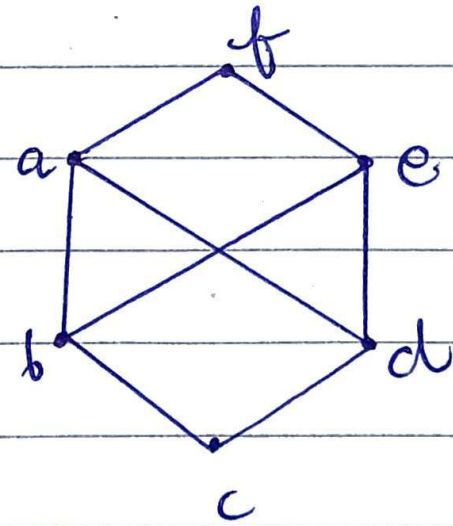
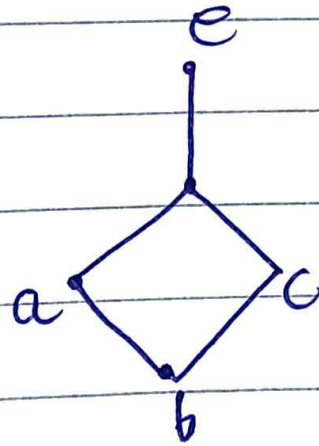
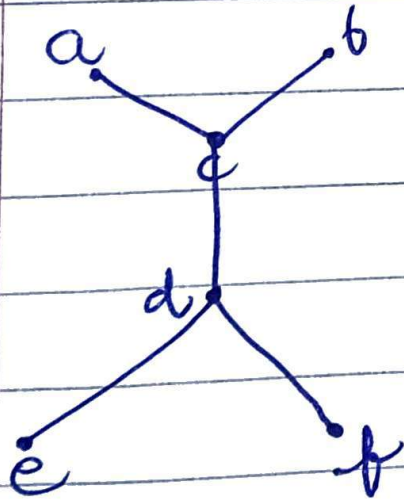
$$a \vee b = \text{lub}\{a, b\}$$

↳ join

$$a \wedge b = \text{glb}$$

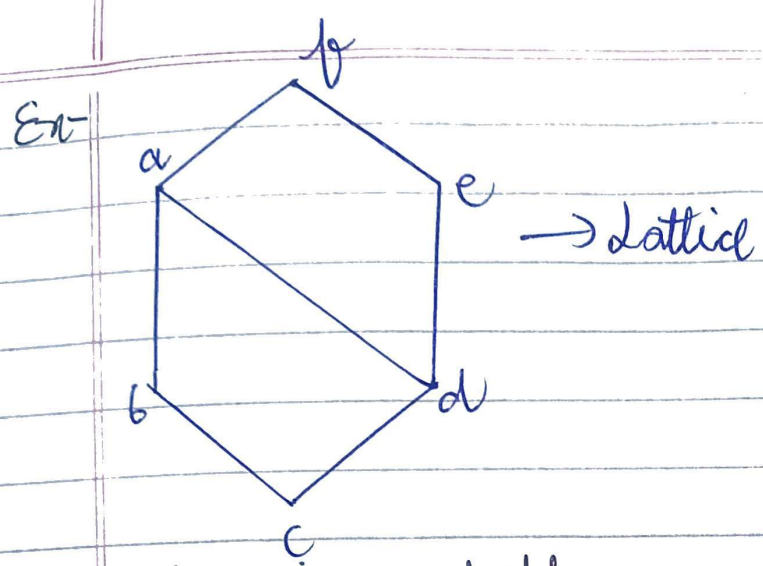
↳ meet

Ex-



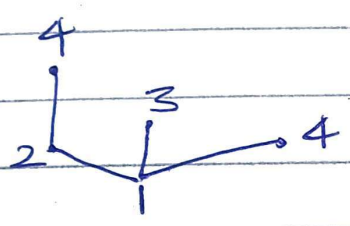
Not a lattice because join of  $(b, d)$  is not unique.

$(b \vee d)$  and  $(a \wedge e)$  is not unique.



Ex-  $(\mathbb{Z}^+, 1)$  or  $(\mathbb{N}, 1)$   
 $a, b \in \mathbb{Z}^+$   
 $a \vee b = \text{lcm} \{a, b\}$   
 $a \wedge b = \text{gcd} \{a, b\}$

Q Determine whether a poset:  $(\{1, 2, 3, 4, 5\}, 1)$



Meet exists but  $(2 \vee 3)$  does not exist. So, it is not a lattice.

Q  $(\{1, 2, 4, 8\}, 1)$



|                |                  |                  |                  |
|----------------|------------------|------------------|------------------|
| $8 \vee 4 = 8$ | $4 \vee 2 = 4$   | $8 \wedge 2 = 2$ | $2 \wedge 1 = 1$ |
| $8 \vee 2 = 8$ | $2 \vee 1 = 2$   | $8 \wedge 1 = 1$ |                  |
| $8 \vee 1 = 8$ | $8 \wedge 4 = 4$ | $4 \wedge 2 = 2$ |                  |

It is a lattice.

Theorem: In any Poset  $(P, \leq)$ , if meets and joins exist, they satisfy the following rules for  $x, y, z \in P$ .

- L<sub>1</sub>:  $x \wedge x = x$  and  $x \vee x = x$
- L<sub>2</sub>:  $x \wedge y = y \wedge x$  and  $x \vee y = y \vee x$
- L<sub>3</sub>:  $(x \wedge y) \wedge z = x \wedge (y \wedge z)$  and  $(x \vee y) \vee z = x \vee (y \vee z)$
- L<sub>4</sub>:  $x \wedge (x \vee y) = x$  and  $x \vee (x \wedge y) = x$

★ Distributive laws does not hold for every lattice.  
 i.e.  $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$   
 and  $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$

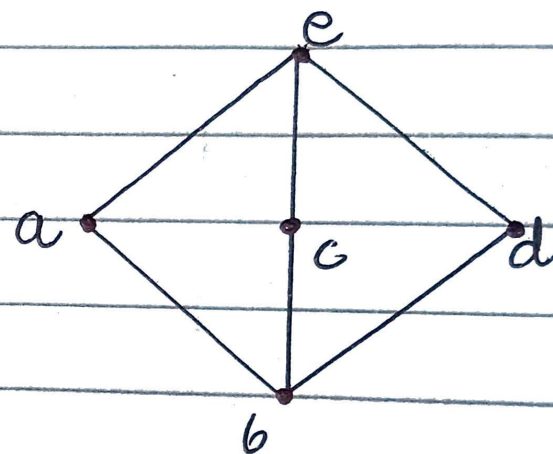


Ex- The following lattice do not satisfy the distributive law:

$$\begin{aligned} &\Rightarrow a \wedge (c \vee d) \\ &= a \wedge e \\ &= a \end{aligned}$$

$$\begin{aligned} &\Rightarrow (a \wedge c) \vee (a \wedge d) \\ &= b \vee b \\ &= b \end{aligned}$$

$$a \neq b$$

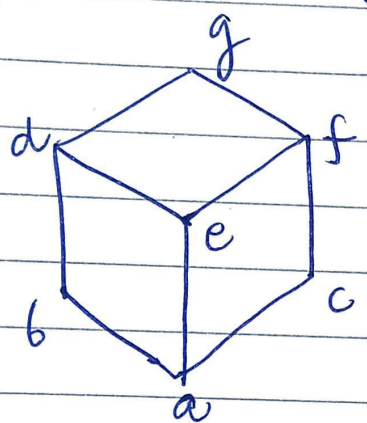


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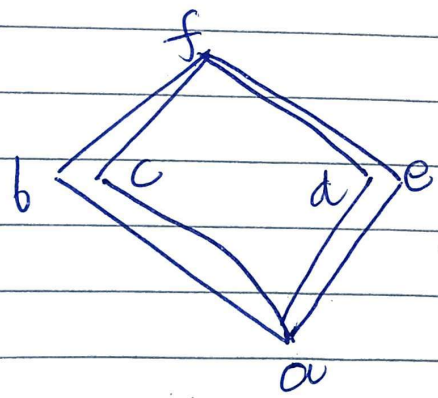
Distributive lattice: Complement of each element is unique.

Let  $I \rightarrow$  greatest element  
 $0 \rightarrow$  least element

Complement: If  $a \wedge x = 0$  &  $a \vee x = I$   
 $a' = x$  or  $x' = a$



$a' = g$  &  $g' = a$   
 $b \vee c = g, b \wedge c = a \Rightarrow b' = c$   
 $e \vee g = g \Rightarrow \therefore e'$  DNE  
 $c \wedge g = e \neq a$



$a' = f$  &  $f' = a$   
 $c' = b, d, e$  &  $c' = b, d, e$   
 $d' = b, c, e$  &  $e' = b, c, d$

Theorem: In distributive lattice  $L$ , a given element  $x$  can have at most one complement.

Proof: Let  $x$  have 2 complements in lattice, say  $y$  &  $z$ .

$\Rightarrow$

|                  |   |                  |
|------------------|---|------------------|
| $x \vee y = I$   | , | $x \vee z = I$   |
| $x \wedge y = 0$ |   | $x \wedge z = 0$ |



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$$\begin{aligned} y &= y \wedge (x \vee y) \\ &= y \wedge (x \vee z) = (y \wedge x) \vee (y \wedge z) \\ &= (x \wedge z) \vee (y \wedge z) = (x \vee y) \wedge z = (x \vee z) \wedge z \\ &= z \end{aligned}$$